VITISH 2024

(SIH Internal Hackathon)

TITLE PAGE

- **Problem Statement ID –25030**
- **Problem Statement Title- AI Based Crop Recommendation for Farmers**
- **Theme- Agriculture, FoodTech & Rural Development**
- **PS Category- Software**
- **Team ID- VIT-229**

IDEA TITLE

Idea/Solution

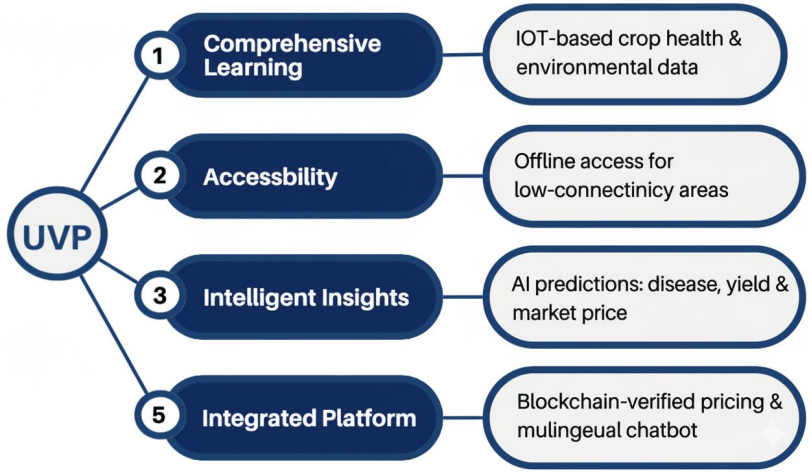
Our solution, Krisideep, is a comprehensive platform designed to empower farmers with real-time, personalized, and actionable guidance for crop management. It integrates:

- **Sensors** track **soil health**, **moisture**, **temperature**, and key **environmental factors**, allowing farmers to **monitor fields offline** and always access **critical data** for timely decisions.
- **Smart machine learning models** predict **crop diseases**, estimate **yield**, recommend **optimal crops**, and forecast **market prices**, helping farmers plan effectively and reduce losses.
- **Secure and transparent market price updates** through **blockchain** ensure reliable information and **fair pricing** for produce.
- **Mobile app** provides **live monitoring**, **predictions**, and **alerts** in **local languages**, making the platform **accessible** and **user-friendly** for farmers.
- **Interactive chatbot** offers **guidance** and answers in **local languages**, supporting farmers in real-time.
- **Live web dashboard** mirrors **app features**, displaying **crop data**, **market trends**, **recommendations**, and **FAQs**.
- Farmers receive **real-time alerts** and **notifications** about **crop health**, **market fluctuations**, and critical events like **price spikes or drops**.
- **Offline accessibility** ensures **core features** remain functional in **low-connectivity areas**, empowering farmers anywhere.

Problem resolution

- Provides farmers with **AI-based crop recommendations**, ensuring better productivity and profitability.
- Overcomes **language and knowledge barriers** through localized advice and intuitive interfaces.

Unique Value Proposition(UVP)



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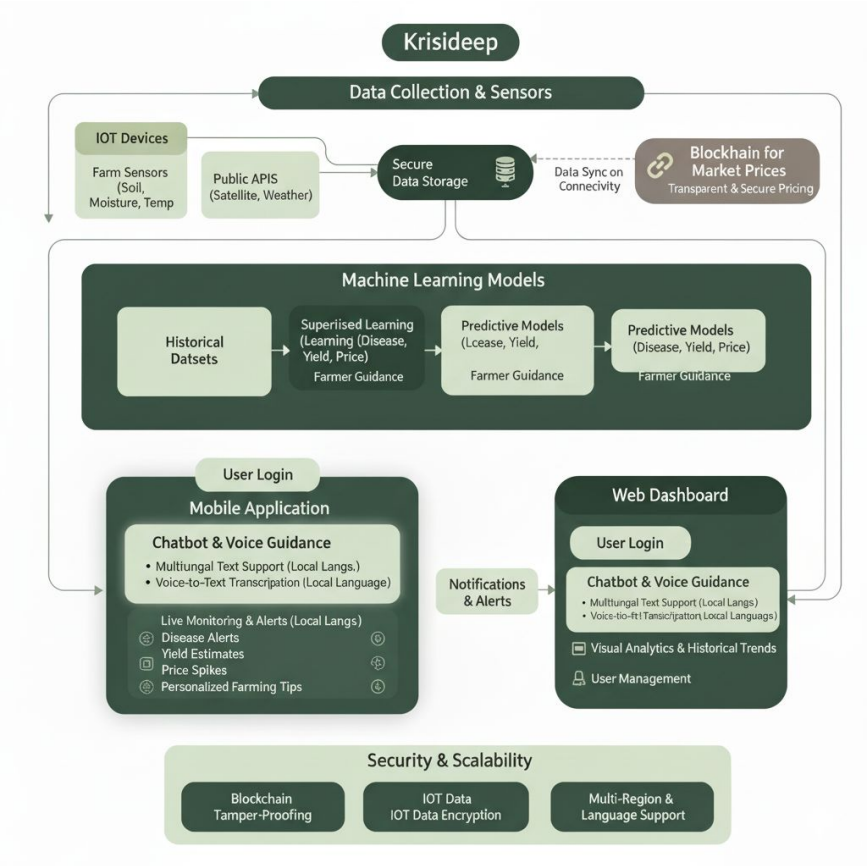
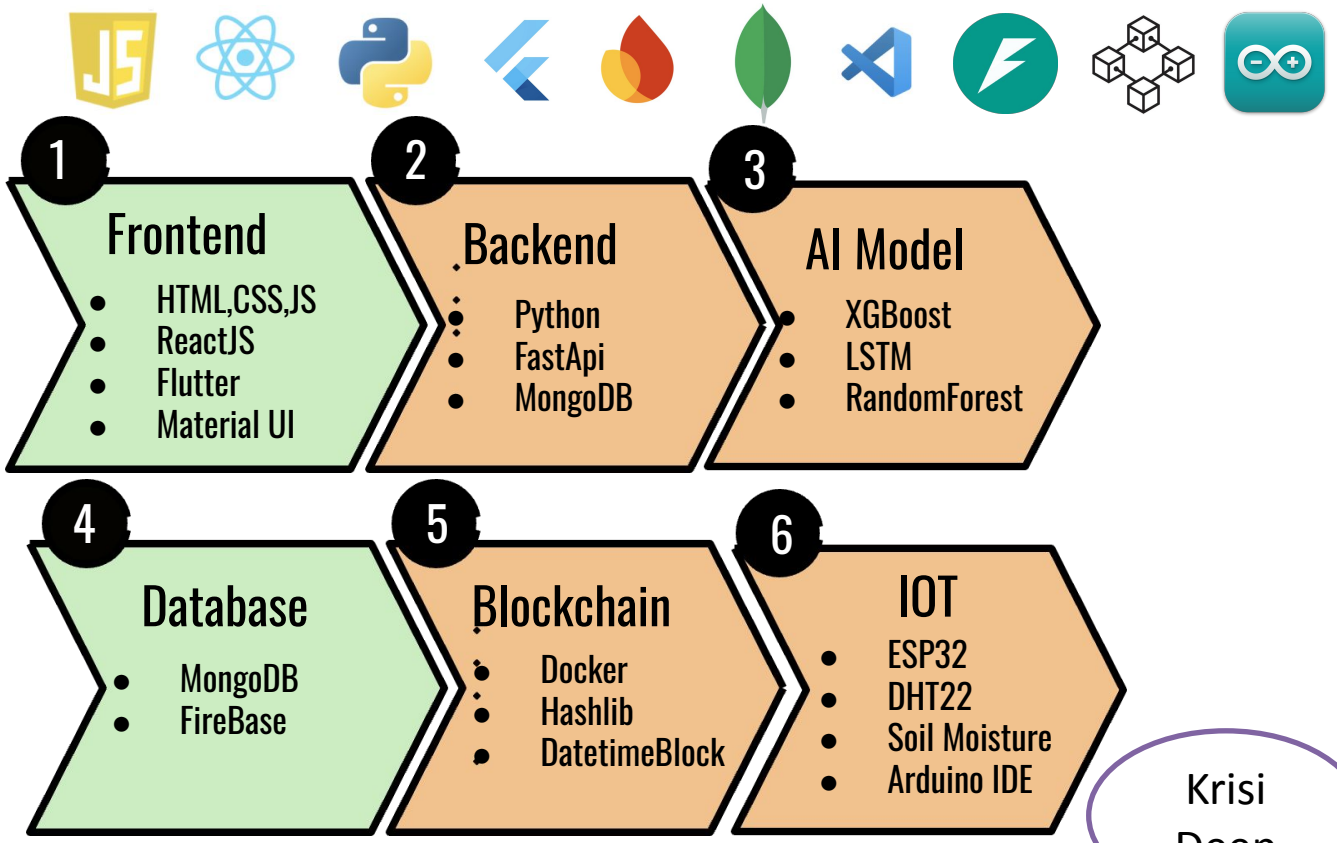
TECHNICAL APPROACH

Methodology



Tech Stack



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FEASIBILITY AND VIABILITY

Analysis of Feasibility

- **Technical Feasibility: IoT sensors** for **soil and environmental monitoring**, **AI/ML models** for **crop prediction**, **blockchain** for **secure pricing**, **mobile and web apps** for **seamless farmer access**.
- **Operational Feasibility: Expert team**, **cloud infrastructure**, and **multi-language support** ensure **scalability** and **reliable growth**.

Potential Challenges and Risks

- **Technical Risks: Complex integration** of **IoT, AI/ML, blockchain**, and **mobile/web systems**; ensuring **smooth, user-friendly experience**.
- **Operational Risks:** Maintaining **accurate and updated crop, soil, and market data**; **scaling system** for multiple regions; encouraging **farmer adoption**.

Strategies for Overcoming Challenges

- **Technical Solutions: Agile development**, **cloud scaling**, and **performance testing** to ensure **smooth integration** of IoT, AI/ML, and blockchain systems.
- **Data Optimization:** Use **large, diverse datasets** to improve **AI/ML prediction accuracy** and provide **reliable crop recommendations**.
- **Operational Solutions: Partnerships**, **farmer feedback**, **multi-language & offline support**.

Business Model

E-Commerce: In-app sales of **seeds, fertilizers, tools, and essential farm products**.

Subscriptions: Premium access to **AI-based crop recommendations, disease alerts, and personalized guidance**.

Advertisements & Sponsorships: Revenue from **agri-brands, equipment makers, and govt schemes**.

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IMPACT AND BENEFITS

Potential impact on the target audience

Educational Value: Provides **real-time AI-driven insights** on crops, soil health, and market trends, **helping farmers learn and make better decisions**.

Inclusivity: **Multi-language support, offline access**, and **user-friendly interface** ensure all farmers, including those in remote areas, can use the platform.

Economic Empowerment: Enables **data-driven crop planning** and **fair market pricing via blockchain**, helping farmers **increase yield and income**.

Environmental Awareness: Promotes **sustainable and eco-friendly farming**, **soil conservation**, and **efficient use of water and resources**.

Wellness & Productivity: Encourages **better farm management practices**, improving **crop health and overall productivity**.

Benefits of the solution

Social: Builds a **community of farmers and agricultural learners**, promotes **knowledge sharing**, and preserves **traditional and local farming practices**.

Economic: Helps farmers **increase yield and income** through **data-driven crop planning**, supports **fair pricing**, and strengthens **local agricultural markets**.

Health Advantages: Improves **farmers' productivity and crop quality**, indirectly supporting **food security** and **well-being** through healthier agricultural practices.

Educational: Provides **real-time AI-driven insights**, **knowledge on crop management**, and boosts **learning** for farmers.

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RESEARCH AND REFERENCES

Government Reports:

- **ICAR (Indian Council of Agricultural Research)** – Reports on smart farming and digital agriculture in India.
- **Ministry of Agriculture & Farmers Welfare** – "Digital Agriculture Mission 2021–2025".

Case Studies:

- **"AI in Indian Agriculture" – NITI Aayog & IBM, 2019** (crop yield prediction & advisory systems).
- **IoT in Precision Farming – Springer, 2020**.

Web Sources:

- **ICAR Official Website: www.icar.org.in**
- **Weather Api: https://openweathermap.org/api**

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